

MOHAMED SHAMIL

AI/ML Engineer

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SUMMARY

Final-year BSc (Hons) Artificial Intelligence undergraduate focused on building agentic AI systems, LLM-based agents, and production-ready backend services. Delivered production multi-agent systems during a 6-month AI/ML internship using MCP, RAG, and LLM architectures to automate multi-step workflows. Currently leading a final-year research project on hybrid LLM–HRM reasoning for interpretable mathematical problem solving, with hands-on experience across FastAPI, Docker, AWS, and PostgreSQL.

EXPERIENCE

CodeGen International (Pvt) Ltd

Mar 2025 – Sep 2025

AI/ML Engineer Intern

Colombo, Sri Lanka

- Delivered production MCP, RAG agent systems that automated multi-step workflows previously handled manually, enabling contextual tool use over domain knowledge.
- Built LLM-based agents with persistent memory and autonomous task execution, allowing multi-step tasks to be planned and completed end to end.
- Improved domain-task reliability through prompt engineering and fine-tuning, aligning model behavior with production workflow requirements.
- Developed FastAPI and PostgreSQL backend services for agent state management and conversation persistence across live sessions.
- Containerized and deployed AI services to AWS with Docker and WhatsApp Business API integration, bringing agentic automation to end users.
- Tools & Technologies:** Python, FastAPI, LangChain, MCP, RAG, LLMs, WhatsApp Business API, PostgreSQL, Docker, AWS.

RESEARCH

Hybrid Architecture of LLM and HRM for Multi-Step Mathematical Problem Solving

2025 – Present

Final-Year Research Project – University of Moratuwa

Advisor: Dr. Romesh Thanuja

- Designed an end-to-end hybrid reasoning pipeline that combines an LLM parser, a graph-based bridge encoder, an HRM core with Adaptive Computation Time halting, and an LLM explanation module for interpretable multi-step mathematical reasoning.
- Converted free-form math word problems into structured dependency graphs with intermediate arithmetic traces, enabling step-wise reasoning and auditable outputs.
- Implemented the HRM high- and low-level transformer modules, graph encoder, digit-head decoder, and PyTorch training loop; trained on GSM8K (7,473 train / 1,319 test).
- Targeting at least 40% exact-match accuracy against standalone LLM and HRM baselines, with interpretability evaluated through per-step decoded values, halt-step diagnostics, and LLM-as-judge explanation coherence scoring.
- Stack:** PyTorch, HRM, Graph Attention Networks, NVIDIA NIM.

PROJECTS

AI-Assisted Carbon Footprint Monitoring System | Scikit-learn, React, Node.js, Spring Boot, Nov 2023 - Jul 2024 GCP | GitHub

- Developed, in a Sysco Labs-mentored team, an ML system that analyzes CO2e emissions across the food supply chain to surface sustainability insights, training Scikit-learn models on cleaned and feature-engineered emissions data.
- Built backend microservices for vendor and product management and interactive data-visualization components for the emissions dashboard.

AI-Driven Multi-Agent Hospital Management System | Python, FastAPI, LangGraph, MCP, 2025 FastMCP, Docker, AWS | GitHub

- Architected a multi-agent hospital automation system that coordinates patient admission, bed assignment, and staff allocation through specialized LLM agents orchestrated with **LangGraph**.
- Exposed hospital operations as AI-accessible tools by implementing the Model Context Protocol (**MCP**) with FastMCP, giving agents a standardized, typed interface to core system functions.
- Built the backend as FastAPI, SQLAlchemy REST services and automated build, test, and deployment to AWS through a GitHub Actions CI/CD pipeline.

WhatsApp Sales Agent - AI Conversational Commerce <i>Python, Google Gemini, Flask, Phi Agent Framework</i> GitHub	2025
- Built an autonomous conversational sales agent on Google Gemini that drives product discovery and recommendations through natural-language chat on WhatsApp. - Engineered a conversation-memory layer - user profiling, message classification, and session persistence - so the agent holds context across multi-turn buying conversations. - Extended the agent with a real-time currency-converter tool for international pricing and deployed it via a Flask webhook server integrated with the Meta WhatsApp Business API.	
ATS Resume Tracking System <i>Python, Streamlit, Google Gemini API</i> GitHub	2024
- Built an LLM-powered application that evaluates resumes against job descriptions - computing match scores, flagging missing keywords, and generating targeted improvement suggestions. - Deployed the tool on Streamlit Community Cloud as a self-service web app for live resume analysis.	
Fuzzy-Enhanced Real-Time Place Recognition & Weapon Detection <i>Python, YOLOv8, Fuzzy Logic, Places365, OpenCV</i> GitHub	2024
Built a real-time surveillance system that fuses YOLOv8 object detection with a fuzzy-logic layer to sustain weapon- and place-recognition reliability under uncertain conditions - varying lighting, camera angles, and occlusions.	
Obesity Classification Prediction <i>Python, Scikit-learn, XGBoost, LightGBM</i> GitHub	2024
Built and benchmarked gradient-boosting classifiers (XGBoost, LightGBM, Decision Trees) to predict obesity levels from medical data, using hyperparameter tuning to select the strongest-performing model.	
Autonomous Chessboard - Hardware/Software Integration <i>Embedded Systems</i>	2022 - 2023
Co-built an AI-assisted autonomous chessboard, delivering the real-time LCD move-display subsystem and contributing to hardware-software integration and testing.	

EDUCATION

University of Moratuwa	2022 - 2026
<i>BSc (Hons) in Artificial Intelligence - CGPA: 3.30 / 4.00</i>	<i>Colombo, Sri Lanka</i>
Relevant coursework: Machine Learning, Deep Learning, Natural Language Processing, Computer Vision, Reinforcement Learning, Data Science, and Algorithms.	
Kekunagolla National School	2018 - 2020
<i>G.C.E. Advanced Level (Physical Science) - Mathematics A, Chemistry A, Physics B</i>	<i>Kurunegala, Sri Lanka</i>

TECHNICAL SKILLS

Agentic AI & LLMs: MCP, LangChain, LangGraph, RAG, multi-agent orchestration, Phi/Agno Agent Framework, prompt engineering, fine-tuning, LLM APIs (Gemini, OpenAI)
Machine Learning & Deep Learning: PyTorch, Scikit-learn, XGBoost, LightGBM, Transformers, Hugging Face, Computer Vision (YOLOv8, OpenCV)
Programming Languages: Python, Java, C, JavaScript
Backend & APIs: FastAPI, Flask, Spring Boot, RESTful APIs, SQLAlchemy
Data & Cloud: PostgreSQL, MySQL, MSSQL, AWS, GCP, Docker, CI/CD (GitHub Actions)
Tools: Git, GitHub, VS Code, IntelliJ IDEA, Jira, Trello

CERTIFICATIONS

Python for Beginners - University of Moratuwa	Python Intermediate	Introduction to Generative AI - Coursera	
Introduction to LLMs	Prompt Engineering	Introduction to Java	Java Intermediate

LANGUAGES

English (Fluent) · Tamil (Native) · Sinhala (Fluent)

LEADERSHIP & INVOLVEMENT

- IEEE Student Member - University of Moratuwa.
- Company Coordinator, Career Fair 2024 - University of Moratuwa.
- Participant, MoraXtreme 8.0 & CodeRush 2023 - inter-university competitive programming and hackathon competitions.

REFERENCES

Dr. Thilini Piyatilake - Senior Lecturer, University of Moratuwa · thilinis@uom.lk

W.M.R.M. Wijesuriya - Lecturer, University of Moratuwa · wijesuriyar@uom.lk

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